EN.553.762 Nonlinear Optimization 2

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Contents

Chapter 1

Linear Programs

Definition 1.0.1. A linear program (LP) minimizes or maximizes a linear function

$$c^T x = \sum_{i=1}^{n} c_i x_i = \langle c, x \rangle$$

over a feasible region defined by linear inequalities $a_i^T x \leq b_i \forall i = 1, ..., m$:

We can also format this as:

$$\begin{cases} \min \max c^T x \\ s.t. \quad Ax \leq b \quad \text{(elementwise)} \end{cases}$$

where

- $A = \text{row}(a_m^T) = \text{col}(A_n)$
- $Ax = \text{col}(a_m^T x) = A_1 x_1 + ... + A_n x_n$
- n is the decision variable dimensions $x \in \mathbb{R}^n$
- m is the number of constraints $b \in \mathbb{R}^m$

Useful terms to know:

1. $\{x | a_i^T \leq b_i\}$ for fixed i is called a *half space*
2. $\mathcal{P} = \{x | a_i^T \leq b_i \forall i = 1, ..., m\}$ is called a *polyhedron*
3. If additional $\mathcal{P}$ is bounded, then it is called a *polytope*
4. We say a LP is *infeasible* if no x satisfies $Ax \leq b$, denoted by

$$\begin{cases} \min(\max) c^T x = \pm\infty \\ s.t. \quad Ax \leq b \end{cases}$$

5. We say a LP is *unbounded* if $\exists$ a sequence $x^{(i)} \in \mathbb{R}^n$ such that $\lim_{i \to \infty} c^T x^{(i)} = -\infty$.
6. We say $x \in \mathbb{R}^n$ is a *minimizer* (or maximizer) if $Ax \leq b$ and $c^T x^* \leq c^T x \forall x \text{s.t.} Ax \leq b$ (or $c^T x^* \geq c^T x \forall x \text{s.t.} Ax \leq b$)

7. Any $x \in \mathbb{R}^n$ is called a *solution*
    - x with $Ax \leq b$ is called a *feasible solution*
    - x minimizing/maximizing is called an *optimal solution*

Theorem 1.0.1. *Every LP is infeasible, unbounded, or has an optimal solution*

Proof. We prove this by counter-example. Consider a nonlinear program

$$\arg\min e^x \text{s.t.} x \in \mathbb{R}$$

. This is actually equivalent to taking an infinum, and results in non-attainment (We will worry about this for NLP but not LP) $\square$

"Standard Form" LPs are

$$\begin{cases} \arg\min c^T x \quad \text{(always minimize)} \\ \text{s.t.} Ax = b \text{(equality constraints related to A)} \\ \quad x \geq 0 \text{(simple inequality constraints)} \end{cases}$$

Theorem 1.0.2. *Every LP can be equivalently rewritten in standard form*

Example 1: (Knapsack Problem) Suppose you have goods $i = 1, ..., n$ and each has three attributes $[w_i, v_i, p_i]$ for weight, volume, and payoff. Given weights and volume upper bounds W, V, you aim to maximize your payoff

$$\begin{cases} \arg\max p^T x = \sum p_i x_i \\ \text{s.t.} w^T x \leq W \\ \quad v^T x \leq V \\ \quad x \geq 0 \end{cases} \iff \begin{cases} \arg\min -p^T x \\ \text{s.t.} w^T x + s_1 \leq W \\ \quad v^T x + s_2 \leq V \\ \quad xs \geq 0 \end{cases}$$

Example 2: (Transportation Problem) Suppose we are shipping goods from supply centers $i = 1, ..., m$ to demand centers $j = 1, ..., n$. Each supply center has s_i supply and each demand center has d_j demand. Shipping cost from i to j is c_{ij} per unit

$$\begin{cases} \arg\min \sum_{ij} c_{ij} x_{ij} \\ \text{s.t.} \sum_j x_{ij} = s_i \forall i \\ \quad \sum_i x_{ij} = d_j \forall j \\ x \geq 0 \end{cases}$$

1.1 Extreme Points

Recall a typical picture of an LP's feasible region is $\mathcal{P} = \{x | Ax \leq b\}$ and the equivalent standard form is $\mathcal{P} = \{x | Ax = b, x \geq 0\}$, that is x is in a subspace and a non-negative orthant. Standard form gives a nice property that the feasible region is closed, i.e. $\{x | x_0 + \lambda d \forall \lambda \in \mathbb{R}\} \subsetneq \mathcal{P}$. We will also refer to extreme points as *corners*. There are three definitions:

- Geometric: we say $x \in \mathcal{P}$ is an *extreme point* if no $y, z \in \mathcal{P}$ with $x = \lambda y + (1-\lambda)z, \lambda \in [0,1]$ exists
- Optimization: we say $x \in \mathcal{P}$ is a *vertex* if some c has $c^T x < c^T y \forall y \in \mathcal{P}$
- Algebraic: we say $x \in \mathcal{P}$ is a *basic feasible solution (BFS)* if there exists a linearly independent a_i with $a_i^T x = b_i$ (implying x is the unique solution with those tight constraints)

Theorem 1.1.1. *For any polyhedron $\mathcal{P} = \{x | Ax \leq b\}$, a point $x \in \mathbb{R}^n$ is an extreme point iff it is a vertex and iff it is a BFS.*

Proof. We will prove this by showing Vertex $\implies$ Extreme Point $\implies$ BFS $\implies$ Vertex. Suppose x is a vertex. Consider any $y, z \in \mathcal{P}$. Now

$$\lambda(c^T x < c^T y) + (1-\lambda)(c^T x < c^T z) = c^T x \leq c^T(\lambda y + (1-\lambda)z)$$

$$\implies x \neq \lambda y + (1-\lambda) z \text{for any} \lambda \in [0,1]$$

Suppose x is not a BFS, Let $I \subseteq \{1, ..., m\}$ such that $a_i^T x = b_i$ ($i \notin I, a_i^T x < b_i$). Then $\{a_i\}_{i \in I}$ does not contain linearly independent vectors

$$\implies \exists d \neq 0 \quad a_i^T d = 0 \quad \forall i \in I$$

$$\implies a_i^T(x \pm \varepsilon d) = a_i^T x = b_i \quad \forall i \in I$$

$$\implies x \pm \varepsilon d \text{is feasible for some } \varepsilon \text{as small as we want}$$

Since $x = \frac{1}{2}((x + \varepsilon d) + (x - \varepsilon d))$, x is not an extreme point. Now suppose that x is BFS with I defined as before. Consider $c = -\sum_{i \in I} a_i$. All $y \in \mathcal{P}$ have

$$c^T y = -\sum_{i \in I} a_i^T y > -\sum_{i \in I} b_i = -\sum_{i \in I} a_i^T x = c^T x$$

BFS $\iff$ n linear independent $a_i^T x = b_i$ where x is an n-dimensional vector $\iff$ x is the only solution to $a_i^T y = b_i$ □

Recall a standard form polyhedron with m equalities and n more inequalities is $\mathcal{P} = \{x | Ax = b, x \geq 0\}$ (need $m < n$, assume A has independent rows).
Pick a *Basis* $B = \{B(1), ..., B(m)\} \subseteq \{1, ..., n\}$ and complement $\bar{B} = \{1, ..., n\}$. Let $x_B = \vec{x}_{B(n)}$, $c_B = \vec{c}_{B(m)}$, $x_{\bar{B}} = \vec{x}_{B(n \times m)}$, $c_{\bar{B}} = \vec{c}_{B(n \times m)}$, $A_B = \text{col}(A_{B(m)})$, and $A_{\bar{B}} = \text{row}(A_{\bar{B}(n \times m)})$.
A BFS comes from forcing $x_{\bar{B}} = 0$

$$\begin{cases} Ax = b \\ x_{\bar{B}} = 0 \end{cases} \iff \begin{cases} A_B x_B + A_{\bar{B}} x_{\bar{B}} = b \\ x_{\bar{B}} = 0 \end{cases}$$

if and only if

$$\begin{cases} A_B x_B = b \\ x_{\bar{B}} = 0 \end{cases} \implies \begin{cases} x_B = A_B^{-1} b \\ x_{\bar{B}} = 0 \end{cases}$$

Lemma 1.1.1. *Every nonempty standard form polyhedron has a BFS*

Consider a standard form LP and recall that x^* is a minimizer if

$$\begin{cases} Ax^* = b, x^* \geq 0 \\ c^T x^* \leq c^T y \forall y \in \mathcal{P} \end{cases}$$

Theorem 1.1.2. *If an LP has a minimizer, then some BFS is a minimizer.*

Proof. Let x^* be a minimizer of $c^T x$ over $\mathcal{P}$. Define $Q = \{x | Ax = b, c^T x = c^T x^*, x \geq 0\}$. This means there must exist a x^* that is a BFS of Q, and we cannot write x^* as an average of minimizers (elements of Q). Consider a BFS x^* with basis B. This implies that a unique x^* solves

$$\begin{cases} Ax = b \\ x_{barB} = 0 \end{cases}$$

□

Consider a BFS $\bar{x}$ with basis B. Try increasing some $x_j = \varepsilon, j \in \bar{B}$.
Let $\tilde{x}$ uniquely solve

$$\begin{cases} Ax = b \\ x_j = \varepsilon \\ x_{\bar{B}} = 0 \end{cases} \iff \begin{cases} \tilde{x}_B = A_B^{-1} b - A_B^{-1} A_j \varepsilon \\ \tilde{x}_j = \varepsilon \\ \tilde{x}_{\bar{B}} = 0 \end{cases}$$

Primal feasibilitity is $x_B = A_B^{-1} b \geq 0$ and dual feasibility is $c - A^T y = c - A^T(A_B^{-1} c_B) \geq 0$

$$c^T \tilde{x} = \underbrace{c_B^T A_B^{-1} b}_{c^T \tilde{x}} - \underbrace{c_B^T A_B^{-1} A_j \epsilon + c_j \epsilon}_{\text{The reduced cost of j in basis B: } \bar{c}_j}$$

Theorem 1.1.3. *Consider a BFS $\bar{x}$ with B basis and $\bar{c}$ reduced costs. (i) if $\bar{c} \geq 0$, then $\bar{x}$ is a minimizer, and (ii) if $\bar{x}$ is a minimizer and nondegenerate, then $\bar{c} \geq 0$.*

Proof of (i). Suppose $\bar{c} \geq 0$. Consider any $x \in P = \{x \mid Ax = b,\ x \geq 0\}$.
We want to show:

$$c^T \bar{x} \leq c^T x.$$

Let

$$d = x - \bar{x}.$$

Since $Ax = b$ and $A\bar{x} = b$, we have

$$Ad = 0.$$

Partitioning into basic and nonbasic components:

$$A_B d_B + A_{\bar{B}} d_{\bar{B}} = 0.$$

Thus,

$$d_B = -A_B^{-1} A_{\bar{B}} d_{\bar{B}} = -\sum_{j \in \bar{B}} A_B^{-1}(A_j d_j).$$

Now,

$$c^T(x-\bar{x}) = c^T d = c_B^T d_B + c_{\bar{B}}^T d_{\bar{B}}.$$

Substitute for d_B:

$$= c_B^T \left(-\sum_{j\in\bar{B}} A_B^{-1} A_j d_j \right) + \sum_{j\in\bar{B}} c_j d_j.$$

$$= -\sum_{j\in\bar{B}} \left(c_B^T A_B^{-1} A_j - c_j \right) d_j.$$

$$= \sum_{j\in\bar{B}} \bar{c}_j\, d_j.$$

Since $x_j \geq 0$ and $\bar{x}_j = 0$ for $j \in \bar{B}$, we have $d_j \geq 0$, and by assumption $\bar{c}_j \geq 0$. Therefore,

$$\sum_{j\in\bar{B}} \bar{c}_j\, d_j \geq 0.$$

Hence,

$$c^T(x-\bar{x}) \geq 0 \quad \Rightarrow \quad c^T\bar{x} \leq c^T x.$$

□

Consider a primal standard form LP:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Write

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix}.$$

Pick multipliers $y \in \mathbb{R}^m$. Then

$$\sum_{i=1}^{m} y_i\,(a_i^T x = b_i) \quad \Longleftrightarrow \quad y^T A x = y^T b.$$

Ensure that

$$y^T A \leq c^T \quad \text{(elementwise)}.$$

Since $x \geq 0$, we obtain

$$c^T x \geq y^T b.$$

The strongest lower bound is given by the dual LP:

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y \leq c. \end{aligned}$$

1.2 Duality

Theorem 1.2.1. *(Weak Duality) For any A, b, c,*

$$\begin{array}{rl} \min & c^T x \\ s.t. & Ax = b \\ & x \geq 0 \end{array} \geq \begin{array}{rl} \max & b^T y \\ s.t. & A^T y \leq c. \end{array}$$

Recall for a basis B, the reduced costs are

$$\bar{c}^T = c^T - c_B^T A_B^{-1} A \geq 0.$$

Define

$$y^T = c_B^T A_B^{-1}.$$

Then

$$A^T y \leq c.$$

Duality allow for several important things in optimization:

- Optimizes proofs
- Variational changes (aka shadow prices) via derivative w.r.t. b
- Lagrange multipliers as a penalty for infeasibility
- Changing points (Fenchel) and linear functions (gradients)

Theorem 1.2.2 (Strong Duality)**.** *For any (A, b, c), if at least one of the primal or dual LP is feasible, then*

$$p^* = \begin{array}{rl} \min & c^T x \\ s.t. & Ax = b \\ & x \geq 0 \end{array} = \begin{array}{rl} \max & b^T y \\ s.t. & A^T y \leq c \end{array} = d^*.$$

Proof. By weak duality,

$$p^* \geq d^*.$$

If the primal is unbounded, then $p^* = -\infty$, and the dual is infeasible, so $d^* = -\infty$. Similarly, if the dual is unbounded, then the primal is infeasible, and

$$p^* = \infty = d^*.$$

□

Proof. WLOG assume the primal is feasible and not unbounded, so an optimal solution x^* exists.

We want to find a dual feasible y^* such that

$$c^T x^* = b^T y^*.$$

Construct a basis B such that:

- The associated primal solution $\bar{x}$ is feasible:

$$A\bar{x} = b, \quad \bar{x}_{\bar{B}} = 0, \quad \bar{x}_B = A_B^{-1} b \geq 0.$$

- The associated dual solution
$$y = A_B^{-T} c_B$$
is feasible.

Dual feasibility follows since the reduced costs satisfy

$$\bar{c}^T = c^T - c_B^T A_B^{-1} A \geq 0 \quad \iff \quad A^T y \leq c.$$

Finally,
$$c^T \bar{x} = c_B^T \bar{x}_B = c_B^T A_B^{-1} b = b^T A_B^{-T} c_B = b^T y.$$
Thus $c^T \bar{x} = b^T y$, completing the proof. □

1.3 The Simplex Method

The core step is *pivoting*. Given a BFS $\bar{x}$ (nondegenerate) with basis B, so that

$$\bar{x}_B = A_B^{-1} b \geq 0, \quad \bar{x}_{\bar{B}} = 0,$$

fix $j \in \bar{B}$ and consider $\tilde{x}(\epsilon)$ defined by

$$\tilde{x}(\epsilon) = \bar{x} + \epsilon d,$$

where the direction d is given by
$$d = \begin{bmatrix} -A_B^{-1} A_j \\ 1 \\ 0 \end{bmatrix}.$$

Then $\tilde{x}(\epsilon)$ uniquely solves

$$Ax = b, \quad x_j = \epsilon, \quad x_{\bar{B} \setminus \{j\}} = 0.$$

1.3.1 Unbounded Case

If $d \geq 0$, then for all $\epsilon \geq 0$,
$$\tilde{x}(\epsilon) = \bar{x} + \epsilon d \geq 0,$$
and by construction,
$$A\tilde{x}(\epsilon) = b.$$
Thus $\tilde{x}(\epsilon)$ is feasible for all $\epsilon \geq 0$.

If $\bar{c}_j < 0$, then
$$c^T \tilde{x}(\epsilon) = c^T \bar{x} + \bar{c}_j \epsilon \to -\infty \quad \text{as } \epsilon \to \infty.$$

Hence the problem is unbounded.

1.3.2 Bounded Case

Suppose $\bar{c}_j < 0$ but $d \not\geq 0$. Then there exists some i such that

$$d_i < 0.$$

Feasibility requires

$$\tilde{x}_i(\epsilon) = \bar{x}_i + \epsilon d_i \geq 0,$$

which holds if and only if

$$\epsilon \leq \frac{\bar{x}_i}{-d_i}.$$

Define the step size

$$\epsilon^* = \min \left\{ \frac{\bar{x}_i}{-d_i} \;\middle|\; d_i < 0 \right\}.$$

Let

$$i \in \arg\min \left\{ \frac{\bar{x}_i}{-d_i} \;\middle|\; d_i < 0 \right\}.$$

Then $\tilde{x}(\epsilon^*)$ is feasible and defines the next BFS.

Lemma 1.3.1. *Let $B' = B \cup \{j\} \setminus \{i\}$. Then B' is a basis for $\tilde{x}(\epsilon^*)$. In particular, $\tilde{x}(\epsilon^*)$ is a BFS.*

Proof. Clearly, $\tilde{x}(\epsilon^*)$ satisfies

$$Ax = b, \quad x_{\bar{B}'} = 0.$$

Thus we need to show that $A_{B'}$ is invertible.

Note that $A_{B'}$ is obtained from A_B by replacing the i-th column with A_j. Let k be the position of i in B. Then

$$A_{B'} = \begin{bmatrix} A_{B(1)} & \cdots & A_j & \cdots & A_{B(m)} \end{bmatrix} = A_B + (A_j - A_i)e_k^T.$$

By the Sherman–Morrison formula, $A_{B'}$ is invertible if and only if

$$1 + e_k^T A_B^{-1}(A_j - A_i) \neq 0.$$

Compute:

$$e_k^T A_B^{-1} A_j = -d_k \quad \text{(from the definition of } d\text{)},$$

and

$$e_k^T A_B^{-1} A_i = e_k^T e_k = 1.$$

Hence,

$$1 + e_k^T A_B^{-1}(A_j - A_i) = 1 + (-d_k - 1) = -d_k.$$

Since $d_k \neq 0$ (indeed $d_k < 0$ by construction), we conclude that $A_{B'}$ is invertible. $\square$

Before the 1980s, people used the Simplex Method, and it could be exponential time in the number of systems to be solved. Afterwards, Interior point methods were invented which solved these systems in polynomial time. Nowadays, we use First-Order approaches which instead of solving a system, it does a ton of matrix-vector multiplications

1.4 A Bit on LPs

The simplex method assumes that a basic feasible solution (BFS) is given.

Consider the standard form LP:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

To obtain an initial BFS, introduce slack variables and solve the Phase I problem:

$$\begin{aligned} \min \quad & \sum_i s_i \\ \text{s.t.} \quad & Ax + Is = b \\ & x, s \geq 0, \end{aligned}$$

where we assume $b \geq 0$. Then $(x, s) = (0, b)$ is a BFS.

Phase I and Phase II

- **Phase I:** Find a feasible solution by minimizing the sum of slack variables.
- **Phase II:** Use the BFS from Phase I to solve the original LP.

Adding constraints to the primal can break primal feasibility, while adding variables to the dual does not affect dual feasibility.

1.5 Modeling Choices

Consider the set

$$P = \{x \mid \|x\|_1 \leq 1\} = \left\{ x \;\middle|\; \sum_{i=1}^{n} |x_i| \leq 1 \right\}.$$

This polytope has:

- 2^n faces corresponding to inequalities

$$(\pm 1, \pm 1, \ldots, \pm 1)^T x \leq 1,$$

- $2n$ BFS given by vectors of the form

$$(0, \ldots, 0, \pm 1, 0, \ldots, 0).$$

Moreover,

$$P = \operatorname{conv}\left(\{\pm e_i\}_{i=1}^{n}\right) = \left\{ \sum_i \lambda_i x_i \;\middle|\; \sum_i \lambda_i = 1,\; \lambda_i \geq 0 \right\}.$$

Quadratic Formulation and Lifting Consider the problem:

$$\begin{aligned} \max \quad & x^T A x = \sum_{i,j} A_{ij} x_i x_j \\ \text{s.t.} \quad & x \in \{\pm 1\}^n. \end{aligned}$$

This can be written using matrix inner products:

$$\max \quad \langle A, X \rangle \quad \text{s.t.} \quad X = xx^T,\ x \in \{\pm 1\}^n,$$

where

$$\langle A, X \rangle = \operatorname{tr}(A^T X) = \sum_{i,j} A_{ij} X_{ij}.$$

Relaxing this, we obtain:

$$\max \quad \langle A, X \rangle \quad \text{s.t.} \quad X \in \operatorname{conv}\{xx^T \mid x \in \{\pm 1\}^n\}.$$

This leads to semidefinite programming relaxations, which can often recover good approximations (e.g., ~ 0.878 for Max-Cut).

Chapter 2

Beyond Linearity

We consider variables x living in a Euclidean, finite-dimensional space E equipped with an inner product $\langle \cdot, \cdot \rangle$ and norm
$$\|x\|^2 = \langle x, x \rangle.$$

Examples:
$$E = \mathbb{R}^n, \quad \langle x, y \rangle = x^T y,$$
$$E = \mathbb{R}^{n \times n}, \quad \langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

We are interested in problems of the form

$$\begin{array}{rl} \min & f(x) \\ \text{s.t.} & x \in Q, \end{array}$$

or equivalently,
$$\begin{array}{rl} \min & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \quad i = 1, \ldots, m, \end{array}$$
where $f, g_i : E \to \mathbb{R} \cup \{+\infty\}$.

The domain of a function $h : E \to \mathbb{R} \cup \{+\infty\}$ is

$$\operatorname{dom}(h) = \{x \in E \mid h(x) < +\infty\}.$$

2.1 Calculus

We say h is (Fréchet) differentiable at $x \in \operatorname{dom}(h)$ if there exists $\nabla h(x) \in E$ such that

$$\lim_{\Delta \to 0} \frac{h(x + \Delta) - \Big(h(x) + \langle \nabla h(x), \Delta \rangle\Big)}{\|\Delta\|} = 0.$$

The vector $\nabla h(x)$ is called the gradient.

If ∇h exists everywhere in $\operatorname{dom}(h)$ and is continuous, then $h \in C^1$.

We say h is twice (Fréchet) differentiable at x if there exists a linear operator

$$\nabla^2 h(x) : E \to E$$

such that
$$\lim_{\Delta \to 0} \frac{\|\nabla h(x+\Delta) - (\nabla h(x) + \nabla^2 h(x)(\Delta))\|}{\|\Delta\|} = 0.$$
The operator $\nabla^2 h(x)$ is called the Hessian. If it exists and is continuous, then $h \in C^2$.

2.2 Convexity

A function f is linear if
$$f(\lambda x) = \lambda f(x), \quad f(x+y) = f(x) + f(y), \quad \forall x, y \in E,\ \lambda \in \mathbb{R}.$$
It is affine if
$$f(\lambda x + (1-\lambda)y) = \lambda f(x) + (1-\lambda) f(y), \quad \forall x, y \in E,\ \lambda \in [0,1].$$
A function f is convex if
$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y), \quad \forall x, y \in E,\ \lambda \in [0,1].$$
It is strictly convex if
$$f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda) f(y), \quad \forall x \neq y,\ \lambda \in (0,1).$$
A function is concave if the inequality is reversed.

A set $Q \subseteq E$ is convex if
$$\lambda x + (1-\lambda) y \in Q, \quad \forall x, y \in Q,\ \lambda \in [0,1].$$

Example

$$f(x) = |x| - 1$$
is convex but not strictly convex.

2.3 Families of Optimization Problems

2.3.1 Quadratic Programs (QP)

Let $E = \mathbb{R}^n$. Consider $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $H \in \mathbb{R}^{n \times n}$ symmetric positive semidefinite.

We study:
$$\begin{array}{ll} \min & c^T x + \frac{1}{2} x^T H x \\ \text{s.t.} & Ax = b \\ & x \geq 0. \end{array}$$
If $H \succeq 0$, then:
$$v^T H v \geq 0 \quad \forall v \in \mathbb{R}^n \quad \Longleftrightarrow \quad H = G^T G,$$
and the objective is convex.

Example 1 (Projection) Let $P = \{x \mid Ax = b,\ x \geq 0\}$. Given $\bar{x} \notin P$, the projection onto P solves:

$$\min_{x \in P} \tfrac{1}{2}\|x - \bar{x}\|_2^2.$$

Expanding,

$$\tfrac{1}{2}\|x\|^2 - \bar{x}^T x + \tfrac{1}{2}\|\bar{x}\|^2,$$

which is a QP with $H = I$, $c = -\bar{x}$ (up to constants).

Example 2 (LASSO)

$$\begin{aligned} \min \quad & \tfrac{1}{2}\|Ax - b\|_2^2 \\ \text{s.t.} \quad & \|x\|_1 \leq \lambda. \end{aligned}$$

Equivalently,

$$\tfrac{1}{2}x^T A^T A x - b^T A x + \tfrac{1}{2}\|b\|^2,$$

with $H = A^T A$.

2.3.2 Second-Order Cone Programs (SOCP)

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & \|A_i x - b_i\|_2 \leq f_i^T x + d_i, \quad i = 1, \dots, k. \end{aligned}$$

LPs are included by setting $A_i = 0$, $b_i = 0$.
QPs can be embedded via epigraph reformulation:

$$\min t \quad \text{s.t.} \quad t \geq c^T x + \tfrac{1}{2}x^T H x.$$

Key representation:

$$\|y\|_2 \leq t \quad \Longleftrightarrow \quad (y, t) \in \mathcal{K}_{\text{soc}}.$$

Example (LASSO revisited)

$$\min \|x\|_1 \quad \text{s.t.} \quad \|Ax - b\|_2 \leq \gamma,$$

which can be written as an SOCP using standard transformations.

2.3.3 Semidefinite Programs (SDP)

Let $E = \mathbb{S}^n$ be the space of symmetric matrices with inner product

$$\langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

$$\begin{aligned} \min \quad & \langle C, X \rangle \\ \text{s.t.} \quad & \mathcal{A}(X) = b \\ & X \succeq 0, \end{aligned}$$

where $\mathcal{A} : \mathbb{S}^n \to \mathbb{R}^m$ is linear.
$X \succeq 0$ means X is positive semidefinite.

Connections

- If X is diagonal, SDP reduces to LP.
- SOCP constraints satisfy:

$$\|x\|_2 \le t \quad \Longleftrightarrow \quad \begin{bmatrix} tI & x \\ x^T & t \end{bmatrix} \succeq 0.$$

Example: Max-Cut Relaxation

The Max-Cut problem:

$$\min x^T A x \quad \text{s.t.} \quad x \in \{\pm 1\}^n.$$

Lift to matrix form:

$$X = xx^T.$$

Relaxation:

$$\begin{aligned} \min \quad & \langle A, X \rangle \\ \text{s.t.} \quad & \operatorname{diag}(X) = \mathbf{1} \\ & X \succeq 0, \end{aligned}$$

dropping the rank constraint.

This is the Goemans–Williamson SDP relaxation.

2.3.4 Conic Programs

A conic program is

$$\begin{aligned} \min \quad & \langle c, x \rangle \\ \text{s.t.} \quad & Ax = b \\ & x \in K, \end{aligned}$$

for some $c \in E$, where $A : E \to \mathbb{R}^m$ is linear, and K is a closed, convex cone in E.

The cone of copositive matrices:

$$K^{\text{cop}} = \{X \in \mathbb{R}^{n \times n} \mid v^T X v \ge 0 \ \ \forall v \ge 0\}.$$

Checking $X \in K^{\text{cop}}$ is NP-hard.

(Solve with interior point methods $\to$ tells us which K are tractable.)

2.3.5 Convex Programs

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \le 0, \quad i = 1, \ldots, m, \end{aligned}$$

where f, g_i are closed, convex, proper (c.c.p.).

f is closed if

$$\liminf_{x' \to x} f(x') = f(x), \quad \forall x \in E.$$

f is proper if

$$\operatorname{dom}(f) = \{x \mid f(x) \in \mathbb{R}\} \text{ is nonempty.}$$

Given any closed, convex, nonempty feasible region

$$S = \{x \mid Ax = b,\ x \in K\},$$

claim:

$$\operatorname{dist}_S(x) = \inf\{\|x - x'\| \mid x' \in S\}$$

is c.c.p.

$$\begin{array}{ll} \min & \langle c, x\rangle \\ \text{s.t.} & x \in S \end{array} \iff \begin{array}{ll} \min & \langle c, x\rangle \\ \text{s.t.} & \operatorname{dist}_S(x) \le 0. \end{array}$$

Epigraph Reformulation

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & g_i(x) \le 0 \end{array} = \min_{x \in E} h(x),$$

where

$$h(x) = \begin{cases} f(x) & \text{if } x \text{ is feasible,} \\ +\infty & \text{otherwise.} \end{cases}$$

$$= \begin{array}{ll} \min & t \\ \text{s.t.} & (x, t) \in \operatorname{epi}(h) \end{array}$$

where

$$\operatorname{epi}(h) = \{(x, t) \mid h(x) \le t\}.$$

$$= \begin{array}{ll} \min & t \\ \text{s.t.} & (x, t, z) \in K^{\text{epi}} \\ & z = 1 \end{array}$$

where

$$K^{\text{epi}} = \left\{(x, t, z) \;\middle|\; \left(\frac{x}{z}, \frac{t}{z}\right) \in \operatorname{epi}(h)\right\}.$$

Thus:

- linear objective
- linear constraints
- one cone constraint

$$\implies \begin{array}{ll} \min & \langle c, x\rangle \\ \text{s.t.} & x \in K \\ & Ax = b. \end{array}$$

2.4 Optimality Conditions

Consider

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q \subseteq E. \end{aligned}$$

Definitions

$\bar{x} \in E$ is a *local minimizer* if

$$f(x) \geq f(\bar{x}) \quad \forall x \text{ in a neighborhood of } \bar{x} \text{ and feasible.}$$

$\bar{x} \in E$ is a *global minimizer* if

$$f(x) \geq f(\bar{x}) \quad \forall x \in Q.$$

Directional Derivative

Define the directional derivative of f at $\bar{x}$ in direction d:

$$f'(\bar{x}; d) = \lim_{t \to 0^+} \frac{f(\bar{x} + td) - f(\bar{x})}{t}.$$

If $f \in C^1$, then

$$f'(\bar{x}; d) = \langle \nabla f(\bar{x}), d \rangle.$$

Example For $f(x) = |x|$ at $\bar{x} = 0$:

$$f'(0; 1) = 1, \quad f'(0; -1) = -1.$$

Normal Cone

Define the normal cone of Q at $\bar{x}$:

$$N_Q(\bar{x}) = \{d \mid \langle d, x - \bar{x} \rangle \leq 0 \ \ \forall x \in Q\}.$$

2.4.1 First-Order Necessary Condition

Theorem 2.4.1. *Suppose Q is closed and convex and $\bar{x} \in Q$ is a local minimizer of f over Q.*

If for any $x \in Q$, $f'(\bar{x}; x - \bar{x})$ exists, then

$$f'(\bar{x}; x - \bar{x}) \geq 0.$$

If f is differentiable at $\bar{x}$, then

$$\langle \nabla f(\bar{x}), x - \bar{x} \rangle \geq 0 \quad \forall x \in Q,$$

equivalently,

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

Proof. Suppose to the contrary that for some $x \in Q$,

$$f'(\bar{x}; x - \bar{x}) < 0.$$

Then for some $0 < t < \delta$,

$$\frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t} < 0 \Rightarrow f(\bar{x} + t(x - \bar{x})) < f(\bar{x}).$$

But

$$\bar{x} + t(x - \bar{x}) = tx + (1 - t)\bar{x} \in Q$$

by convexity, contradicting local optimality. □

2.4.2 First-Order Sufficient Condition

Theorem 2.4.2. *Suppose f and Q are both closed and convex.*

If for all $x \in Q$,

$$f'(\bar{x}; x - \bar{x}) \geq 0$$

for some fixed $\bar{x} \in Q$, then $\bar{x}$ is a global minimizer.

In particular, if f is differentiable at $\bar{x}$,

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}) \iff \bar{x} \textit{ is a global minimizer.}$$

Proof. The key claim is that convexity implies

$$t \mapsto \frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t}$$

is nondecreasing for $t > 0$.

Thus for any $x \in Q$,

$$\lim_{t \to 0^+} \frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t} \geq 0.$$

Taking $t = 1$,

$$f(x) - f(\bar{x}) \geq 0 \Rightarrow f(x) \geq f(\bar{x}),$$

so $\bar{x}$ is globally optimal. □

2.5 Algorithms

2.5.1 Frank–Wolfe Method

Given

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in Q, \end{array}$$

approximate by the linearized problem:

$$\begin{array}{ll} \min & f(x_k) + \langle \nabla f(x_k), x - x_k \rangle \\ \text{s.t.} & x \in Q. \end{array}$$

If Q is polyhedral, run simplex or enumerate BFS.
If $Q = B(0, 1)$, the solution is

$$\bar{x} = \frac{-\nabla f(x_k)}{\|\nabla f(x_k)\|_2}.$$

Iteration:

$$\bar{x}_{k+1} = \arg\min_{x \in Q} \langle \nabla f(x_k), x \rangle,$$

$$x_{k+1} = x_k + \beta_k(\bar{x}_{k+1} - x_k).$$

Choose $\beta_k = \frac{1}{\sqrt{k+1}}$ or via line search:

$$x_{k+1} \in \arg\min_{x \in [x_k, \bar{x}_{k+1}]} f(x).$$

Define

$$G_k = \langle \nabla f(x_k), x_k - \bar{x}_{k+1} \rangle \geq 0.$$

Then

$$\begin{aligned} f(x_{k+1}) &= f(x_k + \beta_k(\bar{x}_{k+1} - x_k)) \\ &\leq f(x_k) + \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{L}{2}\|x_{k+1} - x_k\|_2^2 \\ &= f(x_k) - \beta_k G_k + \frac{L}{2}\beta_k^2\|\bar{x}_{k+1} - x_k\|_2^2. \end{aligned}$$

Assuming $\|\bar{x}_{k+1} - x_k\| \leq 2D$:

$$f(x_{k+1}) \leq f(x_k) - \beta_k G_k + 2LD^2\beta_k^2.$$

Summing:

$$f(x_0) - \min_{x \in Q} f(x) \geq \sum_{k=1}^{T} \beta_k G_k - 2LD^2 \sum_{k=1}^{T} \beta_k^2.$$

Thus,

$$\min_k G_k \leq \frac{f(x_0) - p^* + 2LD^2 \log T}{\sqrt{T+1}}.$$

2.5.2 Projected Gradient Descent

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q. \end{aligned}$$

Iteration:

$$x_{k+1} = \text{proj}_Q(x_k - \beta_k \nabla f(x_k)),$$

where

$$\text{proj}_Q(y) = \arg\min_{z \in Q} \frac{1}{2}\|z - y\|^2.$$

Lemma 2.5.1. *If* $z = \mathrm{proj}_Q(y)$, *then*

$$y - z \in N_Q(z).$$

Proof.

$$-\nabla\left(\tfrac{1}{2}\|z-y\|^2\right) \in N_Q(z) \quad \Rightarrow \quad -(z-y) \in N_Q(z).$$

□

Thus,

$$n_k = (x_k - \beta_k \nabla f(x_k)) - x_{k+1} \in N_Q(x_{k+1}),$$

i.e.

$$-\nabla f(x_k) + \frac{1}{\beta_k}(x_k - x_{k+1}) \in N_Q(x_{k+1}).$$

If $x_k \to \bar{x}$, then

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

2.5.3 Proximal Point Method

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q. \end{aligned}$$

Iteration:

$$x_{k+1} \in \arg\min_{x \in Q} \left\{ f(x) + \frac{\rho}{2}\|x - x_k\|^2 \right\}.$$

Optimality condition:

$$-\nabla\left(f(x) + \tfrac{\rho}{2}\|x - x_k\|^2\right)\Big|_{x=x_{k+1}} \in N_Q(x_{k+1}),$$

i.e.

$$-\nabla f(x_{k+1}) - \rho(x_{k+1} - x_k) \in N_Q(x_{k+1}).$$

2.6 Constrained Optimality Conditions

Theorem 2.6.1 (Weierstrass)**.** *For any closed nonempty set* $Q \subseteq E$, *and closed* $f : Q \to \mathbb{R}$ *with bounded level sets*

$$\{x \mid f(x) \le \lambda\},$$

there exists a global minimizer of f *over* Q.

Theorem 2.6.2 (Basic Separating Hyperplane Theorem)**.** *For any closed convex* $Q \subseteq E$, *and* $y \notin Q$, *there exist* $a \in E$, $b \in \mathbb{R}$ *such that*

$$\langle a, y\rangle > b \ge \langle a, x\rangle \quad \forall x \in Q.$$

Proof. Consider $f(x) = \frac{1}{2}\|x - y\|_2^2$. Then f has bounded level sets.

By Weierstrass, there exists $\bar{x}$ minimizing f over Q, hence

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

Since $\nabla f(\bar{x}) = \bar{x} - y$, we get

$$y - \bar{x} \in N_Q(\bar{x}),$$

i.e.

$$\langle y - \bar{x}, x - \bar{x}\rangle \leq 0 \quad \forall x \in Q.$$

Pick $a = y - \bar{x}$ and $b = \langle y - \bar{x}, \bar{x}\rangle$. Then

$$\langle a, y\rangle > b \geq \langle a, x\rangle \quad \forall x \in Q.$$

□

Theorem 2.6.3 (Small Gradients Exist). *If $f : E \to \mathbb{R}$ is C^1 and bounded below, then there exists a sequence $x_k \in E$ such that*

$$\nabla f(x_k) \to 0.$$

Proof. Fix $\epsilon > 0$ and define

$$h(x) = f(x) + \epsilon\|x\|.$$

Then h has bounded level sets, so it attains a global minimizer x_ϵ.

Let $d = -\nabla f(x_\epsilon)$. For small $t > 0$,

$$\frac{f(x_\epsilon + td) - f(x_\epsilon)}{t} = \frac{h(x_\epsilon + td) - h(x_\epsilon)}{t} - \epsilon\frac{\|x_\epsilon + td\| - \|x_\epsilon\|}{t}.$$

Thus

$$\langle \nabla f(x_\epsilon), d\rangle \geq -\epsilon\|d\|.$$

Since $d = -\nabla f(x_\epsilon)$,

$$-\|\nabla f(x_\epsilon)\|^2 \geq -\epsilon\|\nabla f(x_\epsilon)\| \quad \Rightarrow \quad \|\nabla f(x_\epsilon)\| \leq \epsilon.$$

□

Theorem 2.6.4 (Gordan's Theorem of Alternatives). *For any $a_1, \ldots, a_m \in E$, exactly one of the following holds:*

1. $\exists \lambda \geq 0$, $\sum_i \lambda_i = 1$ *such that*

$$\sum_{i=1}^m \lambda_i a_i = 0,$$

2. $\exists x \in E$ *such that*

$$\langle a_i, x\rangle < 0 \quad \forall i.$$

Lemma 2.6.1 (Farkas Lemma). *For any $a_1, \ldots, a_m \in E$, $c \in E$, exactly one holds:*

1. $\exists \mu \geq 0$ *such that*

$$\sum_{i=1}^m \mu_i a_i = c,$$

2. $\exists x \in E$ *such that*

$$\langle a_i, x\rangle \leq 0 \quad \forall i, \quad \langle c, x\rangle > 0.$$

Constrained Optimality Conditions

Consider

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0. \end{aligned}$$

Assume $f, g_i \in C^1$, and let $\bar{x}$ satisfy $g_i(\bar{x}) \leq 0$.
Define the active set:

$$I(\bar{x}) = \{i \mid g_i(\bar{x}) = 0\}.$$

Lagrangian

For $\lambda \geq 0$, define

$$L(x; \lambda) = f(x) + \sum_i \lambda_i g_i(x).$$

We say $\lambda \geq 0$ is a Lagrange multiplier vector at $\bar{x}$ if:

1. $\bar{x}$ is a critical point of $L(\cdot; \lambda)$:
$$\nabla_x L(\bar{x}; \lambda) = 0,$$
2. complementary slackness:
$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

Theorem 2.6.5 (Fritz John)**.** *If $\bar{x}$ is a local minimizer, then there exists $(\lambda_0, \lambda) \in \mathbb{R}_+ \times \mathbb{R}_+^m$, not all zero, such that*

$$\lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$
$$\lambda_i g_i(\bar{x}) = 0.$$

If $\lambda_0 > 0$, then λ/λ_0 is a Lagrange multiplier vector.

To guarantee $\lambda_0 \neq 0$, we require that there is no $\lambda \geq 0$, not all zero, such that

$$\sum_{i \in I(\bar{x})} \lambda_i \nabla g_i(\bar{x}) = 0.$$

By Gordan's theorem, this is equivalent to the existence of $d \in E$ such that

$$\langle \nabla g_i(\bar{x}), d \rangle < 0 \quad \forall i \in I(\bar{x}).$$

Such a d is a descent direction for all active constraints.

Chapter 3

Duality

Theorem 3.0.1 (Fritz–John)**.** *If $\bar{x}$ is a local minimizer of*

$$\min f(x) \quad s.t.\ g_i(x) \leq 0 \ \ \forall i,$$

and f, g_i are differentiable at $\bar{x}$, then there exists $(\lambda_0, \lambda) \in \mathbb{R}_+ \times \mathbb{R}_+^m$, not all zero, such that

$$\lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$

$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

We want to guarantee $\lambda_0 \neq 0$.

$$\iff \nexists \lambda \geq 0,\ \lambda \neq 0 \text{ such that } \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,\ \lambda_i g_i(\bar{x}) = 0.$$

$$\iff \nexists \lambda \geq 0,\ \sum_{i \in I(\bar{x})} \lambda_i = 1,\ \sum_{i \in I(\bar{x})} \lambda_i \nabla g_i(\bar{x}) = 0.$$

$$\iff \exists d \in E \text{ such that } \langle \nabla g_i(\bar{x}), d \rangle < 0 \quad \forall i \in I(\bar{x}).$$

Theorem 3.0.2 (Karush–Kuhn–Tucker)**.** *Under the same conditions as Fritz–John, if MFCQ holds at $\bar{x}$, then there exists $\lambda \geq 0$ such that*

$$\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$

$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

Proof. Fritz–John gives (λ_0, λ). MFCQ implies $\lambda_0 \neq 0$. Normalize to $(1, \lambda/\lambda_0)$. □

3.1 Mangasarian–Fromovitz Constraint Qualification (MFCQ)

Lemma 3.1.1. *For convex $g_i \in C^1$, MFCQ holds at all $\bar{x}$ feasible iff there exists x_s such that*

$$g_i(x_s) < 0 \quad \forall i.$$

Proof. If MFCQ holds at $\bar{x}$, then $\bar{x} + td$ is strictly feasible.

If x_s exists, then for any $\bar{x}$, let $d = x_s - \bar{x}$. For $t < 1$,

$$\frac{g_i(\bar{x} + td) - g_i(\bar{x})}{t} \leq \frac{g_i(x_s) - g_i(\bar{x})}{1} < 0.$$

□

Proof of Fritz–John. Let

$$h(x) = \max\{f(x) - f(\bar{x}),\ g_i(x)\}_{i \in I(\bar{x})}.$$

Then $\bar{x}$ is a local minimizer of h.

Claim:

$$h'(\bar{x}; d) = \max\{\langle \nabla f(\bar{x}), d \rangle,\ \langle \nabla g_i(\bar{x}), d \rangle\}_{i \in I(\bar{x})}.$$

First-order optimality gives

$$h'(\bar{x}; d) \geq 0 \quad \forall d,$$

i.e.

$$\nexists d \text{ such that } \begin{cases} \langle \nabla f(\bar{x}), d \rangle < 0, \\ \langle \nabla g_i(\bar{x}), d \rangle < 0. \end{cases}$$

Thus,

$$\exists \lambda_0, \lambda \geq 0 \text{ such that } \lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0.$$

□

3.2 Lagrangian and Duality

Recall the Lagrangian:

$$L(x; \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) = f(x) + \lambda^T g(x).$$

Claim:

$$\sup_{\lambda \geq 0} L(x; \lambda) = \begin{cases} f(x) & \text{if } g_i(x) \leq 0, \\ +\infty & \text{otherwise.} \end{cases}$$

Primal problem

$$p_* = \min f(x) \quad \text{s.t. } g_i(x) \leq 0 = \min_x \max_{\lambda \geq 0} L(x; \lambda).$$

Dual problem

$$d_* = \max_{\lambda \geq 0} \min_x L(x; \lambda) = \max_{\lambda \geq 0} \Phi(\lambda),$$

where $\Phi(\lambda)$ is the dual function.

Theorem 3.2.1 (Weak Duality)**.** *For any $f, g_i : E \to \mathbb{R} \cup \{+\infty\}$,*

$$p_* \geq d_*.$$

Proof. Let $\lambda^{(i)} \geq 0$ attaining d_*. For any feasible $\bar{x}$,

$$f(\bar{x}) = \sup_{\lambda \geq 0} L(\bar{x}; \lambda) \geq \limsup L(\bar{x}; \lambda^{(i)}) \geq \limsup \inf_x L(x; \lambda^{(i)}) = d_*.$$

□

3.3 Value Function and Strong Duality

Define the value function:

$$v(\Delta) = \min f(x) \quad \text{s.t. } g_i(x) \leq \Delta_i.$$

Theorem 3.3.1 (Lagrangian Strong Duality)**.** *Suppose $v : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ is convex and $v(0) = p_*$. Then*

$$p_* = d_* \quad \iff \quad v \text{ is lower semicontinuous at } 0.$$

A function is lower semicontinuous at $\bar{x}$ if

$$\liminf_{x \to \bar{x}} f(x) \geq f(\bar{x}).$$

Notes: - If all f, g_i are convex, then $v(\cdot)$ is convex. - If there exists x_s with $g_i(x_s) < 0$ (Slater), then v is lower semicontinuous.

Theorem 3.3.2 (Strong Duality with Attainment)**.** *If $f, g_1, \ldots, g_m$ are convex and there exists x_s with $g_i(x_s) < 0$, then*

$$p_* = d_*$$

and the optimum is attained.

(Proof deferred: Fenchel duality)

Example: Linear Programming

$$\min -b^T y \quad \text{s.t. } A^T y - c \leq 0$$

$$= \min_y \max_{x \geq 0} \left(-b^T y + x^T (A^T y - c) \right) \geq \max_{x \geq 0} \min_y (\cdot)$$

$$\Phi(x) = \min_y \left(-b^T y + x^T A^T y - x^T c \right) = -c^T x + \min_y (Ax - b)^T y$$

$$= \begin{cases} -c^T x & Ax = b, \\ -\infty & \text{otherwise.} \end{cases}$$

$$\Rightarrow \max_{x \geq 0} \Phi(x) = \max -c^T x \quad \text{s.t. } Ax = b,\ x \geq 0.$$

3.4 Norm Optimization

Given a norm $\|\cdot\|$, the dual norm is

$$\|v\|_* = \sup_{\|x\|=1} \langle v, x\rangle.$$

$$p_* = \min \|x\| \quad \text{s.t. } Ax \leq b.$$

$$L(x;\lambda) = \|x\| + \lambda^T(Ax - b).$$

$$\Phi(\lambda) = \begin{cases} -\langle b, \lambda\rangle & \|A^T\lambda\|_* \leq 1, \\ +\infty & \text{otherwise.} \end{cases}$$

$$d_* = \max -\langle b, \lambda\rangle \quad \text{s.t. } \|A^T\lambda\|_* \leq 1,\ \lambda \geq 0.$$

3.5 Subgradient Calculus

Consider $f : E \to \mathbb{R} \cup \{+\infty\}$.

Convex:

$$\forall x, y \in E,\ \lambda \in [0,1], \quad f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y).$$

Proper:

$$\operatorname{dom} f = \{x \mid f(x) \in \mathbb{R}\} \neq \emptyset.$$

Closed: f is lower semicontinuous on $\operatorname{dom} f$, i.e.

$$\forall x \in \operatorname{dom} f, \quad \liminf_{x' \to x} f(x') \geq f(x).$$

3.5.1 Subgradients

Definition: We say $\phi \in E$ is a subgradient of f at $\bar{x} \in \operatorname{dom} f$ if

$$\forall x \in E, \quad f(x) \geq f(\bar{x}) + \langle \phi, x - \bar{x}\rangle.$$

Subdifferential:

$$\partial f(\bar{x}) = \{\phi \mid \forall x,\ f(x) \geq f(\bar{x}) + \langle \phi, x - \bar{x}\rangle\}.$$

3.5.2 Basic Properties

If f is C^1 and convex:

$$\partial f(x) = \{\nabla f(x)\}.$$

If

$$f(x) = \max_{i=1,\dots,m} f_i(x), \quad f_i \in C^1 \text{ convex},$$

then

$$\partial f(x) = \left\{ \sum_{i \in I(x)} \lambda_i \nabla f_i(x) \;\middle|\; \sum_{i \in I(x)} \lambda_i = 1,\ \lambda_i \geq 0 \right\},$$

where

$$I(x) = \{i \mid f_i(x) = f(x)\}.$$

Examples

$$f(x) = |x| = \max\{-x, x\}$$

$$\partial f(x) = \begin{cases} \{1\} & x > 0, \\ [-1,1] & x = 0, \\ \{-1\} & x < 0. \end{cases}$$

For a norm $f(x) = \|x\|$:

$$\phi \in \partial f(0) \iff \forall x,\ \|x\| \geq \langle \phi, x \rangle \iff \|\phi\|_* \leq 1.$$

$$\Rightarrow \partial \|\cdot\|(0) = \{\phi \mid \|\phi\|_* \leq 1\}.$$

3.5.3 Directional Derivatives

Lemma 3.5.1.

$$\phi \in \partial f(x) \iff \langle \phi, v \rangle \leq f'(x; v) \quad \forall v \in E.$$

Proof. Let $\phi \in \partial f(x)$.

$$f'(x; v) = \lim_{t \downarrow 0} \frac{f(x+tv) - f(x)}{t} \geq \lim_{t \downarrow 0} \frac{f(x+tv) - f(x)}{t} \geq \langle \phi, v \rangle.$$

□

By the lemma:

$$f'(x; v) \geq \max_{\phi \in \partial f(x)} \langle \phi, v \rangle.$$

We seek a subgradient attaining equality.

Existence of Subgradients

Let $S = \operatorname{epi} f$.

Pick $t_n \downarrow 0$ and define

$$x_n = \bar{x} + t_n v, \quad f_n = f(\bar{x}) + t_n f'(\bar{x}; v).$$

Then

$$f'(\bar{x}; v) \le \frac{f(x_n) - f(\bar{x})}{t_n} \Rightarrow f(x_n) \ge f(\bar{x}) + t_n f'(\bar{x}; v).$$

Thus (x_n, f_n) lies on or below the graph.

By separating hyperplane theorem, there exists (g_n, h_n) such that

$$\langle (g_n, h_n), (x, t) - (x_n, f_n) \rangle \le 0 \quad \forall (x, t) \in \operatorname{epi} f.$$

Rescale so $h_n = -1$.

Then

$$\begin{gathered} \langle (g_n, -1), (x, f(x)) - (x_n, f_n) \rangle \le 0 \\ \iff f(x) \ge f_n + \langle g_n, x - x_n \rangle. \end{gathered}$$

Taking limits,

$$f(x) \ge f(\bar{x}) + \langle \bar{g}, x - \bar{x} \rangle \quad \forall x,$$

so

$$\bar{g} \in \partial f(\bar{x}).$$

Conclusion:

$$\partial f(\bar{x}) \ne \emptyset \quad \text{for all } \bar{x} \in \operatorname{int}(\operatorname{dom} f).$$

3.5.4 Max Formula

Theorem 3.5.1 (Max Formula)**.** *For any closed, convex, proper f with $\bar{x} \in \operatorname{int}(\operatorname{dom} f)$, and any $v \in E$,*

$$f'(\bar{x}; v) = \max_{\phi \in \partial f(\bar{x})} \langle \phi, v \rangle.$$

In particular,

$$\partial f(\bar{x}) \ne \emptyset.$$

3.6 Duality of Points and Subgradients (Fenchel Duality)

Given $\bar{x} \in \operatorname{int}(\operatorname{dom} f)$, ask for $\phi \in \partial f(\bar{x})$.

Reverse question: Given $\phi \in E$, find $\bar{x}$ such that $\phi \in \partial f(\bar{x})$.

Claim:

$$\bar{x} \text{ globally minimizes } f \quad \iff \quad 0 \in \partial f(\bar{x}).$$

Proof.

$$0 \in \partial f(\bar{x}) \iff \forall x,\ f(x) \geq f(\bar{x}) + \langle 0, x - \bar{x} \rangle = f(\bar{x}) \iff \bar{x} \text{ is a global minimizer.}$$

□

Reverse formulation:

$$\begin{aligned} \phi \in \partial f(\bar{x}) &\iff 0 \in \partial(f - \langle \phi, \cdot \rangle)(\bar{x}) \\ &\iff \bar{x} \text{ minimizes } f(x) - \langle \phi, x \rangle \\ &\iff \bar{x} \text{ maximizes } \langle \phi, x \rangle - f(x). \end{aligned}$$

Fenchel Conjugate

This supremum is called the **Fenchel conjugate**:

$$f^*(\phi) = \sup_{x \in E} \{ \langle \phi, x \rangle - f(x) \}.$$

Properties of f^*

(1) If $\bar{x}$ attains $\sup \langle \phi, x \rangle - f(x)$, then $\phi \in \partial f(\bar{x})$.

(2) If $\phi \in \partial f(\bar{x})$, then $\bar{x}$ attains the supremum.

(3) For all x, ϕ:

$$f^*(\phi) + f(x) \geq \langle \phi, x \rangle.$$

(Fenchel–Young inequality)

$$f^*(\phi) + f(x) \geq \langle \phi, x \rangle,$$

with equality iff $\phi \in \partial f(x)$.

(4) f^* is convex (regardless of f).

(5) f^* is closed.

(6) $f^{**} = f$ iff f is closed and convex.

Examples

(1) Linear function

$$f(x) = \langle c, x \rangle \quad \Rightarrow \quad f^*(\phi) = \begin{cases} 0 & \phi = c, \\ +\infty & \text{otherwise.} \end{cases}$$

(2) Indicator at a point

$$f(x) = \begin{cases} 0 & x = c, \\ +\infty & \text{else,} \end{cases} \quad \Rightarrow \quad f^*(\phi) = \langle c, \phi \rangle.$$

(3) Nonnegative indicator

$$f(x) = \begin{cases} 0 & x \geq 0, \\ +\infty & \text{else,} \end{cases} \quad \Rightarrow \quad f^*(\phi) = \begin{cases} 0 & \phi \leq 0, \\ +\infty & \text{else.} \end{cases}$$

(4) Power function

$$f(x) = \frac{1}{p}|x|^p \quad \Longleftrightarrow \quad f^*(\phi) = \frac{1}{q}|\phi|^q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

(5)

$$f(x) = \sqrt{1+x^2} \quad \Longleftrightarrow \quad f^*(\phi) = \begin{cases} -\sqrt{1-\phi^2} & |\phi| \leq 1, \\ +\infty & \text{else.} \end{cases}$$

Convexity of f^*

$$\lambda f^*(\phi) + (1-\lambda) f^*(\psi) \geq \langle \lambda\phi + (1-\lambda)\psi, x\rangle - f(x)$$
$$\Rightarrow f^*(\lambda\phi + (1-\lambda)\psi) \leq \lambda f^*(\phi) + (1-\lambda) f^*(\psi).$$

Closedness of f^*

$$f^*(\phi_i) \geq \langle \phi_i, x\rangle - f(x)$$
$$\Rightarrow \liminf f^*(\phi_i) \geq \langle \phi, x\rangle - f(x)$$
$$\Rightarrow \liminf f^*(\phi_i) \geq \sup_x(\langle \phi, x\rangle - f(x)) = f^*(\phi).$$

Fenchel Duality

Theorem 3.6.1 (Fenchel Duality)**.** *For $f : E \to (-\infty, \infty]$, $g : Y \to (-\infty, \infty]$, and linear $A : E \to Y$:*

$$p^* = \inf_{x \in E}\{f(x) + g(Ax)\},$$
$$d^* = \sup_{\phi \in Y}\{-f^*(A^T\phi) - g^*(-\phi)\}.$$

In general,

$$p^* \geq d^*.$$

If f, g are convex and

$$0 \in \operatorname{int}(\operatorname{dom} g - A \operatorname{dom} f),$$

then

$$p^* = d^*, \quad d^* \text{ attained.}$$

Proof. By Fenchel–Young:

$$f(x) + f^*(A^T\phi) \geq \langle x, A^T\phi\rangle = \langle Ax, \phi\rangle,$$
$$g(Ax) + g^*(-\phi) \geq \langle Ax, -\phi\rangle.$$

Summing:

$$f(x) + g(Ax) + f^*(A^T\phi) + g^*(-\phi) \geq 0.$$

Thus

$$f(x) + g(Ax) \geq -f^*(A^T\phi) - g^*(-\phi).$$

Taking inf over x and sup over ϕ:

$$p^* \geq d^*.$$

□

Perturbation View

Define

$$h(u) = \inf_x\{f(x) + g(Ax + u)\}, \quad h(0) = p^*.$$

Then h is convex and

$$\operatorname{dom} h = \operatorname{dom} g - A \operatorname{dom} f.$$

There exists $-\phi \in \partial h(0)$, so

$$h(0) \leq h(u) + \langle \phi, u\rangle.$$

$$\Rightarrow h(0) \leq f(x) + g(Ax + u) + \langle \phi, u\rangle$$

$$= f(x) - \langle A^T\phi, x\rangle + g(Ax + u) - \langle \phi, Ax + u\rangle.$$

Taking inf:

$$h(0) \leq -f^*(A^T\phi) - g^*(-\phi) = d^* \leq p^*.$$

3.7 Biconjugates and Lagrange Duality

3.7.1 Stationarity via Subgradients

Recall

$$f^*(\phi) = \sup_{x\in E}\{\langle \phi, x\rangle - f(x)\}.$$

We want:

$$0 \in \partial\Big(f + g \circ A\Big)(\bar{x}).$$

Equivalently,

$$\exists\ \psi \in \partial f(\bar{x}), \quad \phi \in \partial g(A\bar{x}) \quad \text{s.t.} \quad \psi + A^T\phi = 0.$$

Recall the dual:

$$d^* = \sup_\phi\Big(-f^*(A^T\phi) - g^*(-\phi)\Big).$$

3.7.2 Fenchel–Young Symmetry

Suppose $f = f^{**}$.

Then

$$f^*(\phi) + f(\bar{x}) = \langle \phi, \bar{x} \rangle \quad \iff \quad \phi \in \partial f(\bar{x}).$$

Also

$$f^{**}(\bar{x}) + f^*(\phi) = \langle \bar{x}, \phi \rangle \quad \iff \quad \bar{x} \in \partial f^*(\phi).$$

If f, f^* are C^1, then

$$\phi = \nabla f(x) \quad \iff \quad x = \nabla f^*(\phi),$$

so

$$\nabla f^* = (\nabla f)^{-1}.$$

3.7.3 Biconjugate Inequality

Lemma 3.7.1. *For any $h : E \to (-\infty, \infty]$,*

$$h^{**}(x) \leq h(x), \quad \forall x.$$

Proof. By Fenchel–Young:

$$\langle x, \phi \rangle - h^*(\phi) \leq h(x).$$

Taking supremum over ϕ:

$$h^{**}(x) = \sup_{\phi} \{\langle x, \phi \rangle - h^*(\phi)\} \leq h(x).$$

□

3.7.4 Biconjugate as Convex Envelope

The biconjugate h^{**} is the largest closed convex lower bound of h.

$$\operatorname{epi} h^{**} = \operatorname{cl}\Big(\operatorname{conv}(\operatorname{epi} h)\Big).$$

3.7.5 Affine Minorants

Let

$$\alpha(x) = \langle \phi, x \rangle + c$$

be an affine minorant of h if $\alpha(x) \leq h(x)$.

Theorem 3.7.1. *For any $h : E \to (-\infty, \infty]$, the following are equivalent:*

(i) *h is closed and convex*

(ii) $h = h^{**}$

(iii) $h(x) = \sup\{\alpha(x) \mid \alpha \text{ affine minorant of } h\}$

3.7.6 Lagrange Duality via Value Function

Recall the Lagrangian:

$$L(x, \lambda) = f(x) + \sum_i \lambda_i g_i(x).$$

$$p^* = \inf_x \sup_{\lambda \geq 0} L(x, \lambda), \quad d^* = \sup_{\lambda \geq 0} \inf_x L(x, \lambda).$$

Define the value function:

$$v(\Delta) = \inf_x \{f(x) \mid g_i(x) \leq \Delta_i\}.$$

Then:

(1) $p^* = v(0)$

(2) $v^*(-\lambda) = \begin{cases} -\Phi(\lambda) & \lambda \geq 0 \\ +\infty & \text{otherwise} \end{cases}$

(3) $d^* = v^{**}(0)$

Thus weak duality follows from:

$$v^{**} \leq v.$$

3.7.7 Strong Duality via Regularity

If f, g_i are convex, then v is convex.

We need:

$$\liminf_{\Delta \to 0} v(\Delta) \geq v(0)$$

(lower semicontinuity at 0).

This is implied by existence of a strictly feasible point:

$$\exists\ x_s \text{ such that } g_i(x_s) < 0 \quad \forall i.$$

Then:

$$0 \in \operatorname{int}(\operatorname{dom} v) \Rightarrow \exists \phi \in \partial v(0) \Rightarrow v \text{ is l.s.c. at } 0 \Rightarrow v^{**}(0) = v(0).$$

Hence:

$$p^* = d^*.$$

Chapter 4

Methods

4.1 Subgradient and Bundle Methods

Basic Setup

Problem:

$$\min_x f(x), \quad f : E \to \mathbb{R} \text{ convex.}$$

Algorithm idea: At each iteration compute

$$g_k \in \partial f(x_k).$$

(Note: not our choice.)

Example

$$f(x) = \max_i \{\langle a_i, x\rangle + b_i\} + \tfrac{1}{2}\|x\|^2.$$

Then

$$\partial f(x) \ni a_i + x, \quad \text{for any } i \text{ attaining the maximum.}$$

4.1.1 Subgradient Method

$$x_{k+1} = x_k - \alpha_k g_k, \quad g_k \in \partial f(x_k).$$

For $f \notin C^1$, it may happen that

$$f(x_{k+1}) > f(x_k) \quad \text{for all } \alpha_k > 0.$$

Proximal View

$$x_{k+1} = \arg\min_y \left\{ f(y) + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Linearization:

$$f(y) \approx f(x_k) + \langle g_k, y - x_k\rangle.$$

Then

$$x_{k+1} = \arg\min_y \left\{ f(x_k) + \langle g_k, y - x_k \rangle + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Optimality:

$$g_k + \frac{1}{\alpha_k}(y - x_k) = 0 \quad \Longrightarrow \quad y = x_k - \alpha_k g_k.$$

Interpretation:

$$0 \in \partial f(x_k) \quad \text{(approximately)}.$$

4.1.2 Dual Averaging

Define

$$\ell_k(x) = \frac{\sum_{i=1}^k \lambda_i \Big(f(x_i) + \langle g_i, x - x_i \rangle \Big)}{\sum_{i=1}^k \lambda_i}, \quad g_i \in \partial f(x_i).$$

Update:

$$x_{k+1} = \arg\min_y \left\{ \ell_k(y) + \frac{\beta_k}{2\sum \lambda_i}\|y - x_0\|^2 \right\}.$$

This gives:

$$x_{k+1} = x_0 - \frac{\sum \lambda_i g_i}{\beta_k} = x_0 - \sum \alpha_i g_i.$$

Stopping Criterion

$$f(x_k) - \ell_k(x_k) \le \varepsilon, \quad \|\nabla \ell_k(x_k)\| \le \varepsilon.$$

Constrained Case

$$\min f(x) \quad \text{s.t. } x \in S.$$

Projected subgradient method:

$$x_{k+1} = \arg\min_{y \in S} \left\{ f(x_k) + \langle g_k, y - x_k \rangle + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}$$

$$= \mathrm{proj}_S(x_k - \alpha_k g_k).$$

4.1.3 Bundle Methods

Build model m_k:

$$z_{k+1} = \arg\min_y \left\{ m_k(y) + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Descent step:

$$f(z_{k+1}) \le f(x_k) - \text{margin} \Rightarrow x_{k+1} = z_{k+1}.$$

Null step:

$$x_{k+1} = x_k,$$

$$m_{k+1}(y) = \max\left\{ m_k(y), f(z_{k+1}) + \langle g_{k+1}, y - z_{k+1} \rangle \right\}, \quad g_{k+1} \in \partial f(z_{k+1}).$$

Steepest Descent Direction

$$\min_{\|v\|\le 1} f'(x_k; v) = \min_{\|v\|\le 1} \max_{g\in\partial f(x_k)} \langle g, v\rangle.$$

By strong duality:

$$= \max_{g\in\partial f(x_k)} \min_{\|v\|\le 1} \langle g, v\rangle.$$

Inner minimization:

$$\min_{\|v\|\le 1} \langle g, v\rangle = -\|g\|, \quad \text{attained at } v = -\frac{g}{\|g\|}.$$

Thus:

$$= -\min_{g\in\partial f(x_k)} \|g\|.$$

4.1.4 Subgradient Method for Constraints

If x_k feasible:

$$x_{k+1} = x_k - \alpha_k \phi_k, \quad \phi_k \in \partial f(x_k).$$

If infeasible:

$$x_{k+1} = x_k - \alpha_k \phi_k, \quad \phi_k \in \partial g_i(x_k), \quad g_i(x_k) > 0.$$

Goal:

$$f_{\text{best}}^{(k)} = \min_{j\le k,\ x_j \text{ feasible}} f(x_j) \to f(x^*).$$

Step Size Assumptions

$$\alpha_k > 0, \quad \sum \alpha_k = \infty, \quad \sum \alpha_k^2 < \infty, \quad \alpha_k = \frac{1}{k}.$$

Assume:

- f, g_i convex
- bounded subgradients $\|g_k\| \le G$
- Slater point $\tilde{x}$ with $g_i(\tilde{x}) < 0$

Let

$$R \ge \|x_0 - x^*\|,\ \|x_0 - \tilde{x}\|.$$

Convergence Result

$$f_{\text{best}}^{(k)} \to f(x^*).$$

Key Inequality

For feasible x_k:

$$\phi_k^T(x_k - \tilde{x}) \geq f(x_k) - f(\tilde{x}).$$

For infeasible:

$$\phi_k^T(x_k - \tilde{x}) \geq g_i(x_k) - g_i(\tilde{x}) \geq \mu > 0.$$

Thus:

$$\|x_{k+1} - \tilde{x}\|^2 \leq \|x_k - \tilde{x}\|^2 - \delta\alpha_k + G^2\alpha_k^2.$$

Summing:

$$0 \leq R^2 - \delta \sum \alpha_k + G^2 \sum \alpha_k^2.$$

$$\Rightarrow \delta \leq \frac{R^2 + G^2 \sum \alpha_k^2}{\sum \alpha_k} \to 0.$$

Contradiction $\Rightarrow$ convergence.

Lagrangian Interpretation

$$\min_x \max_{\lambda \geq 0} \left\{ f(x) + \sum \lambda_i g_i(x) \right\}.$$

Stopping when:

$$\|\nabla f + \sum \lambda_i \nabla g_i\| \leq \varepsilon.$$

4.2 Penalty and Lagrangian Methods

4.2.1 Penalty Method

$$P_* \approx \min_x \left\{ f(x) + \rho \sum_{i=1}^m \max(0, g_i(x))^2 \right\}.$$

If $f, g_i \in C^1$, then

$$f(x) + \rho \sum_i \max(0, g_i(x))^2 \in C^1.$$

Let

$$\bar{x}(\rho) \in \arg\min \left\{ f(x) + \rho \sum_i \max(0, g_i(x))^2 \right\}.$$

Theorem 4.2.1. *If $\bar{x}(\rho) \to x_*$ as $\rho \to \infty$, then x_* globally minimizes f over $\{g_i(x) \leq 0\}$.*

Proof.

$$f(\bar{x}(\rho)) + \sum \rho \max(0, g_i(\bar{x}(\rho)))^2 \leq P_*, \quad \Rightarrow \quad f(\bar{x}(\rho)) \leq P_*.$$

If for contradiction $g_i(\bar{x}(\rho)) \geq \varepsilon > 0$, then

$$f(\bar{x}(\rho)) + \rho\varepsilon^2 \to \infty,$$

a contradiction.

Thus $\bar{x}(\rho)$ must approach feasibility, and the result follows. □

Exact Penalization

$$\operatorname{dist}_C(x) = \min_{y \in C} \|y - x\|.$$

$$\min f(x) \quad \text{s.t. } x \in C.$$

Assume:

- f is globally M-Lipschitz
- dist_C is tractable

Solve:

$$\min \{f(x) + M' \operatorname{dist}_C(x)\}, \quad M' > M.$$

Theorem 4.2.2. *This problem admits minimizers with $x \in C$.*

Proof. Let $x \notin C$ and $y_i \to C$ with $\|y_i - x\| \to \operatorname{dist}_C(x)$.

$$f(y_i) \leq f(x) + M\|x - y_i\| < f(x) + M'\|x - y_i\|.$$

Taking limits:

$$\lim f(y_i) \leq f(x) + M' \operatorname{dist}_C(x).$$

Since $\operatorname{dist}_C(y_i) = 0$, feasible points dominate. □

4.2.2 Augmented Lagrangian Method

$$\min f(x) \quad \text{s.t. } Ax = b.$$

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^T(Ax - b) + \frac{\rho}{2}\|Ax - b\|^2.$$

$$\min_x \max_\lambda \mathcal{L}_\rho(x, \lambda).$$

Primal-Dual Iteration

$$z_k = \begin{bmatrix} x_k \\ \lambda_k \end{bmatrix}, \quad z_{k+1} = z_k - \alpha_k T(z_k),$$

$$T(z_k) \in \begin{bmatrix} \partial f(x_k) + A^T\lambda_k + \rho A^T(Ax_k - b) \\ b - Ax_k \end{bmatrix}.$$

Convergence

Theorem 4.2.3. *Suppose f is convex, $\|\phi_k\| \leq G$, and iterates are bounded. If*

$$\sum \gamma_k = \infty, \quad \sum \gamma_k^2 < \infty,$$

with $\alpha_k = \gamma_k / \|T(z_k)\|$, then

$$f(x_k) \to f(x_*), \quad \|Ax_k - b\| \to 0.$$

Proof.

$$\|z_{k+1} - z_*\|^2 = \|z_k - z_*\|^2 - 2\gamma_k \frac{T(z_k)^T (z_k - z_*)}{\|T(z_k)\|} + \gamma_k^2.$$

Summing yields boundedness and convergence once we show:

$$T(z_k)^T (z_k - z_*) \geq 0.$$

Compute:

$$T(z_k)^T (z_k - z_*) = \phi_k^T (x_k - x_*) + \lambda_k^T (Ax_k - b) + \rho \|Ax_k - b\|^2 - (Ax_k - b)^T (\lambda_k - \lambda_*).$$

$$= \phi_k^T (x_k - x_*) + \lambda_*^T (Ax_k - b) + \rho \|Ax_k - b\|^2$$

$$\geq f(x_k) - f(x_*) + \lambda_*^T (Ax_k - b) + \rho \|Ax_k - b\|^2 \geq 0.$$

Thus the claim holds and convergence follows. □

4.3 Decomposition Methods

4.3.1 Primal Decomposition

Consider

$$\min_{x_1, x_2, y} f_1(x_1, y) + f_2(x_2, y).$$

Rewrite:

$$= \min_y \left(\inf_{x_1} f_1(x_1, y) \right) + \left(\inf_{x_2} f_2(x_2, y) \right).$$

Exercise 4.3.1. If f_i is convex, then

$$\phi_i(y) = \inf_{x_i} f_i(x_i, y)$$

is convex.

Primal Decomposition Method

Solve in parallel:

- Find x_1 minimizing ϕ_1-problem and $g_1 \in \partial\phi_1(y_k)$
- Find x_2 minimizing ϕ_2-problem and $g_2 \in \partial\phi_2(y_k)$

Update:

$$y_{k+1} = y_k - \alpha_k(g_1 + g_2).$$

At runtime:

$$\text{OPT} \leq f_1(x_1^*, y_k) + f_2(x_2^*, y_k).$$

4.3.2 Dual Decomposition

$$\min_{x_1,x_2,y} f_1(x_1, y) + f_2(x_2, y)$$

$$= \min_{x_1,x_2,y_1,y_2} f_1(x_1, y_1) + f_2(x_2, y_2) \quad \text{s.t. } y_1 = y_2.$$

Lagrangian:

$$= \min_{x_i,y_i} \max_{\lambda} f_1(x_1, y_1) + f_2(x_2, y_2) + \lambda^T(y_1 - y_2).$$

Dual Function

$$\Phi(\lambda) = \underbrace{\min_{x_1,y_1}\left(f_1(x_1, y_1) + \lambda^T y_1\right)}_{\Phi_1(\lambda)} + \underbrace{\min_{x_2,y_2}\left(f_2(x_2, y_2) - \lambda^T y_2\right)}_{\Phi_2(\lambda)}.$$

$$\Phi_1(\lambda) = -(f_1)^*(0, -\lambda), \quad \Phi_2(\lambda) = -(f_2)^*(0, \lambda).$$

If x_i, y_i minimize the subproblems:

$$-y_1 \in \partial\Phi_1(\lambda), \quad y_2 \in \partial\Phi_2(\lambda),$$

$$\Rightarrow y_2 - y_1 \in \partial(-\Phi_1 - \Phi_2)(\lambda).$$

Dual Decomposition Method

Solve in parallel:

$$x_1, y_1 = \arg\min f_1(x_1, y_1) + \lambda_k^T y_1,$$

$$x_2, y_2 = \arg\min f_2(x_2, y_2) - \lambda_k^T y_2.$$

Update:

$$\lambda_{k+1} = \lambda_k - \alpha_k(y_2 - y_1), \quad (y_2 - y_1) \in \partial\Phi(\lambda_k).$$

$$\text{OPT} \geq \text{Dual OPT} \geq \Phi_1(\lambda_k) + \Phi_2(\lambda_k).$$

4.3.3 ADMM (Alternating Direction Method of Multipliers)

$$\min f(x) + g(z) \quad \text{s.t. } Ax + Bz = c.$$

Augmented Lagrangian:

$$\mathcal{L}_\rho(x, y, z) = f(x) + g(z) + y^T(Ax + Bz - c) + \frac{\rho}{2}\|Ax + Bz - c\|^2.$$

Updates:

$$x_{k+1} \in \arg\min_x \mathcal{L}_\rho(x, y_k, z_k),$$

$$z_{k+1} \in \arg\min_z \mathcal{L}_\rho(x_{k+1}, y_k, z),$$

$$y_{k+1} = y_k + \rho(Ax_{k+1} + Bz_{k+1} - c).$$

4.3.4 Applications

4.3.4.1 Consensus Optimization

$$\min_x \sum_i f_i(x) = \min_{x_i, z} \sum_i f_i(x_i) \quad \text{s.t. } x_i = z.$$

ADMM updates:

$$x_i^{k+1} \in \arg\min f_i(x_i) + y_i^{kT}(x_i - z_k) + \frac{\rho}{2}\|x_i - z_k\|^2,$$

$$z^{k+1} = \frac{1}{m}\sum_i \left(x_i^{k+1} + \frac{1}{\rho}y_i^k\right),$$

$$y_i^{k+1} = y_i^k + \rho(x_i^{k+1} - z^{k+1}).$$

4.3.4.2 Exchange Optimization

$$\min \sum_i f_i(x_i) \quad \text{s.t. } \sum_i x_i = 0.$$

4.3.4.3 Example: Power System

Each x_i is production at facility i:

$$f_i(x_i) = \text{cost}.$$

$$\min \sum_i f_i(x_i) \quad \text{s.t. } \sum_i x_i = \text{Target}.$$

$$= \min_x \max_y \sum_i f_i(x_i) + y\left(\sum_i x_i - \text{Target}\right).$$

4.3.5 Network Flow Example

Directed graph with p nodes and n arcs.

Let x_j be flow on arc j:

$$x_j > 0 \text{ forward}, \quad x_j < 0 \text{ reverse}.$$

At node i:

$$s_i > 0 \text{ inflow}, \quad s_i < 0 \text{ outflow}.$$

Flow conservation:

$$Ax + s = 0,$$

where A is the incidence matrix.

Costs

Each arc has cost $\phi_j(x_j)$ (strictly convex).

Example (M/M/1 queue):

$$\phi_j(x_j) = \begin{cases} \frac{x_j}{c_j - x_j} & x_j \in [0, c_j) \\ +\infty & \text{otherwise.} \end{cases}$$

Optimization Problem

$$\min \sum_j \phi_j(x_j) \quad \text{s.t. } Ax + s = 0.$$

Dual Decomposition

$$\mathcal{L}(x, v) = \sum_j \phi_j(x_j) + v^T(Ax + s).$$

$$= v^T s + \sum_j \left(\phi_j(x_j) + (A_j^T v) x_j\right).$$

Let

$$(\Delta v)_j = A_j^T v.$$

Dual function:

$$\Phi(v) = v^T s + \sum_j \inf_{x_j} \left(\phi_j(x_j) + (\Delta v)_j x_j\right).$$

$$= v^T s - \sum_j \sup_{x_j} \left(-(\Delta v)_j x_j - \phi_j(x_j)\right).$$

$$= v^T s - \sum_j \phi_j^*(-(\Delta v)_j).$$

Dual Problem

$$\max_v \left\{ v^T s - \sum_j \phi_j^*(-(\Delta v)_j) \right\}.$$

If ϕ_j strictly convex, then ϕ_j^* has unique maximizer.
Let $x_j^*(v)$ denote it.
Claim.

$$-(Ax^*(\Delta v) + s) \in \partial(-\Phi)(v).$$

4.3.6 Special Case (LP)

$$\min f(x) + g(z) \quad \text{s.t. } Ax = z,$$

$$f(x) = \begin{cases} \langle c, x\rangle & x \geq 0 \\ +\infty & \text{else,} \end{cases} \qquad g(z) = \begin{cases} 0 & z = b \\ +\infty & \text{else.} \end{cases}$$

$$\iff \min \langle c, x\rangle \quad \text{s.t. } Ax = b,\ x \geq 0.$$

4.4 Interior Point and Newton Methods

4.4.1 Barrier Method (LP Sketch)

Consider the linear program:

$$P^\star = \min c^T x \quad \text{s.t. } Ax = b,\ x \geq 0.$$

Introduce barrier:

$$f(x) = \begin{cases} -\sum_{i=1}^n \log(x_i) & x > 0 \\ +\infty & \text{else.} \end{cases}$$

Solve:

$$x_\eta^\star \in \arg\min\ \eta\, c^T x + f(x) \quad \text{s.t. } Ax = b, \quad \eta > 0.$$

Goal:

$$x_\eta^\star \to x^\star \quad \text{as } \eta \to \infty.$$

First-Order Optimality

$$\nabla(\eta c^T x + f(x))(x_\eta^\star) \perp \{x : Ax = b\}$$

$$\Rightarrow \nabla(\eta c^T x + f(x))(x_\eta^\star) = A^T y_\eta^\star.$$

Theorem 4.4.1.

$$b^T y_\eta^\star \geq c^T x_\eta^\star - \frac{n}{\eta}.$$

Thus large η gives near-optimal solutions.

Central Path

As η increases, $x^\star_\eta$ traces the central path.

Barrier Method Algorithm

Given $\eta_0 > 0$ and x_0 approximately solving:

$$\min \eta_0 c^T x + f(x) \quad \text{s.t. } Ax = b,$$

iterate:

$$\eta_{k+1} = \alpha \eta_k, \quad \alpha > 1,$$

$$x_{k+1} \approx \arg\min \eta_{k+1} c^T x + f(x) \quad \text{s.t. } Ax = b,$$

using warm-start from x_k.
Each stage uses a few Newton steps.

Differentiability and Geometry

We work in Euclidean space E with inner product:

$$\langle x, y \rangle = x^T y.$$

For matrices:

$$\langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

Definition 4.4.1. f is (Fréchet) differentiable at x if

$$\lim_{\Delta \to 0} \frac{|f(x+\Delta) - (f(x) + \langle g, \Delta \rangle)|}{\|\Delta\|} = 0.$$

Example: Log-Det Barrier

$$f(X) = \begin{cases} -\ln\det(X) & X \succ 0 \\ +\infty & \text{else.} \end{cases}$$

Theorem 4.4.2.

$$\nabla f(X) = -X^{-1}.$$

Proof. Using eigenvalues λ_i of $X^{-1}\Delta$:

$$f(X+\Delta) - f(X) = -\sum_i \ln(1+\lambda_i),$$

and

$$\langle -X^{-1}, \Delta \rangle = -\sum_i \lambda_i.$$

Using $\ln(1+\lambda) \approx \lambda$ gives result. $\square$

Inner Product Dependence

Theorem 4.4.3. *If f is differentiable under $\langle \cdot, \cdot \rangle$ with gradient g, then under*

$$\langle x, y \rangle_S = \langle x, Sy \rangle,$$

the gradient is $S^{-1}g$.

Proof.

$$f(x+\Delta) - f(x) \approx \langle g, \Delta \rangle = \langle S^{-1}g, \Delta \rangle_S.$$

□

Thus:

Differentiability is inner-product invariant, but gradients are not.

Hessian

Definition 4.4.2. The Hessian $H(x)$ satisfies:

$$\lim_{\Delta \to 0} \frac{\|g(x+\Delta) - (g(x) + H(x)\Delta)\|}{\|\Delta\|} = 0.$$

Example

$$f(X) = -\ln\det(X) \Rightarrow \nabla^2 f(X)[\Delta] = X^{-1}\Delta X^{-1}.$$

4.4.2 Newton's Method

Theorem 4.4.4. *If z minimizes f, then the Newton step $x^+ = x + n(x)$ satisfies:*

$$\|x^+ - z\| \leq \|H(x)^{-1}\| \int_0^1 \|H(x + t(z-x)) - H(x)\| dt \cdot \|x - z\|.$$

Proof. Use $g(z) = 0$ and the fundamental theorem:

$$g(x) - g(z) = \int_0^1 H(x + t(z-x))(z-x)\, dt.$$

Substitute into Newton update. □

Theorem 4.4.5. *If H is Q-Lipschitz, then:*

$$\|x^+ - z\| \leq \|H(x)^{-1}\| \frac{Q}{2} \|x - z\|^2.$$

Thus Newton has quadratic convergence.

4.4.3 Newton with Constraints

Let

$$L = \{x : Ax = b\}.$$

Optimality:

$$-\nabla f(x) \in N_L(x) \quad \Longleftrightarrow \quad -\nabla f(x) = A^T y.$$

Linearization:

$$-\nabla f(x + \Delta) \approx -(\nabla f(x) + \nabla^2 f(x)\Delta) = A^T y.$$

Solve:

$$\min f(x) + \langle g(x), \Delta\rangle + \frac{1}{2}\langle \Delta, H(x)\Delta\rangle \quad \text{s.t. } A(x + \Delta) = b.$$

Projected Form

Let projection onto L:

$$g_L(x) = P_L g(x), \quad H_L(x) = P_L H(x).$$

Theorem 4.4.6. *Newton direction:*

$$n_L(x) = -H_L(x)^{-1} g_L(x).$$

Intrinsic Geometry

Definition 4.4.3. Define intrinsic inner product:

$$\langle y, z\rangle_x = \langle y, H(x)z\rangle.$$

Intrinsic norm:

$$\|y\|_x = \sqrt{\langle y, H(x)y\rangle}.$$

Then:

$$g_x(x) = H(x)^{-1} g(x) = -n(x), \quad H_x(x) = I.$$

4.4.4 Self-Concordance

Definition 4.4.4. Let

$$B_x(y, r) = \{z : \|z - y\|_x \le r\}.$$

f is self-concordant if:

$$B_x(x, 1) \subset \operatorname{dom} f$$

and for $y \in B_x(x, 1)$:

$$1 - \|y - x\|_x \le \frac{\|v\|_y}{\|v\|_x} \le \frac{1}{1 - \|y - x\|_x}.$$

Example

$$f(x) = -\sum_i \log x_i.$$

$$g(x) = \begin{bmatrix} 1/x_1 \\ \vdots \\ 1/x_n \end{bmatrix}, \quad H(x) = \begin{bmatrix} 1/x_1^2 & & \\ & \ddots & \\ & & 1/x_n^2 \end{bmatrix}.$$

For $y \in B_x(x, 1)$:

$$\sum_i \frac{(y_i - x_i)^2}{x_i^2} < 1 \Rightarrow y_i > 0.$$

Also:

$$\|v\|_y^2 \le \|v\|_x^2 \cdot \max_i \frac{x_i^2}{y_i^2} \le \frac{1}{(1 - \|y - x\|_x)^2}.$$

Self-Concordance and Newton Behavior

Recall the intrinsic geometry:

$$\langle y, z\rangle_x = \langle y, H(x)z\rangle, \quad \|y\|_x = \sqrt{\langle y, y\rangle_x}.$$

Then:

$$g_x(x) = H(x)^{-1}g(x) = -n(x), \quad H_x(x) = I.$$

Definition 4.4.5. f is self-concordant (SC) if for all $x \in dom f$:

1. $B_x(x, 1) \subset dom f$
2. For all $y \in B_x(x, 1)$:

$$1 - \|y - x\|_x \le \frac{\|v\|_y}{\|v\|_x} \le \frac{1}{1 - \|y - x\|_x} \quad \forall v \ne 0.$$

Theorem 4.4.7. *f is SC iff for all $y \in B_x(x, 1)$:*

$$\|H_x(y)\|_x,\ \|H_x(y)^{-1}\|_x \le \frac{1}{(1 - \|y - x\|_x)^2}.$$

Equivalently:

$$\|H_x(x) - H_x(y)\|_x \le \frac{1}{(1 - \|y - x\|_x)^2} - 1.$$

Theorem 4.4.8. *If $f \in$ SC and $y \in B_x(x, 1)$, then the second-order model satisfies:*

$$|f(y) - q_x(y)| \le \frac{\|y - x\|_x^3}{3(1 - \|y - x\|_x)}.$$

Proof.

$$f(y) - q_x(y) = \int_0^1 \int_0^t \langle (H_x(x + s(y - x)) - I)(y - x), y - x\rangle_x ds dt.$$

Use SC bounds:

$$\|H_x(x + s(y - x)) - I\|_x \le \frac{1}{(1 - s\|y - x\|_x)^2} - 1.$$

Integrating gives the result. □

Newton Step Contraction

Let:

$$x^+ = x + n(x).$$

Theorem 4.4.9. *If $f \in \mathrm{SC}$ and z is a minimizer with $z \in B_x(x,1)$, then:*

$$\|x^+ - z\|_x \le \frac{\|x-z\|_x^2}{1 - \|x-z\|_x}.$$

Proof. Apply previous Newton bound and SC control on Hessian variation. □

Theorem 4.4.10. *If $f \in \mathrm{SC}$ and $\|n(x)\|_x < 1$, then:*

$$\|n(x^+)\|_{x^+} \le \left(\frac{\|n(x)\|_x}{1 - \|n(x)\|_x}\right)^2.$$

Proof. Use:

$$\|n(x^+)\|_{x^+}^2 = \langle g_x(x^+), H_x(x^+)^{-1} g_x(x^+)\rangle_x,$$

and bound via SC Hessian variation and integral representation of gradients. □

Theorem 4.4.11. *If $f \in \mathrm{SC}$ and*

$$\|n(x)\|_x \le \frac{1}{4},$$

then f has a minimizer $z \in B_x(x,1)$ and:

$$\|z - x^+\|_x \le \frac{3\|n(x)\|_x^2}{(1 - \|n(x)\|_x)^3}.$$

Proof. Using model accuracy:

$$f(y) \ge f(x) - \|n(x)\|_x \|y-x\|_x + \frac{1}{2}\|y-x\|_x^2 - \frac{1}{2}\|y-x\|_x^3.$$

For $\|y - x\|_x \le \frac{1}{3}$:

$$f(y) \ge f(x),$$

so a minimizer exists in the ball. □

4.4.5 Self-Concordant Barriers

Definition 4.4.6. f is a self-concordant barrier (SCB) if:

$$f \in \mathrm{SC} \quad \text{and} \quad \nu_f := \sup_{x \in dom f} \|g_x(x)\|_x^2 < \infty.$$

Example

$$f(x) = -\sum_i \log x_i \quad \Rightarrow \quad \nu_f = n.$$

Closure Properties

Lemma 4.4.1. *If* $f_1, f_2 \in \mathrm{SCB}$*, then:*

$$f_1 + f_2 \in \mathrm{SCB}, \quad \nu_{f_1+f_2} \leq \nu_{f_1} + \nu_{f_2}.$$

Lemma 4.4.2. *If* $f \in \mathrm{SCB}$*, then:*

$$f(Ax - b) \in \mathrm{SCB},$$

with same or smaller ν*.*

4.4.5.1 Barrier Constructions

Polytope

$$\min c^T x \quad \text{s.t. } Ax \leq b, \ \|x\|_2 \leq 1.$$

Barrier:

$$f(x) = -\sum_i \log(b_i - \langle a_i, x\rangle) - \log(1 - \|x\|_2^2).$$

$$\nu_f = m + 1.$$

Positive Orthant

$$x \geq 0 \quad \Rightarrow \quad f(x) = -\sum_i \log x_i, \quad \nu_f = n.$$

PSD Cone

$$X \succeq 0 \quad \Rightarrow \quad f(X) = -\ln\det X, \quad \nu_f = n.$$

If also entrywise positive:

$$f(X) = -\ln\det X - \sum_{i,j} \log X_{ij}, \quad \nu_f = n + n^2.$$

4.4.6 Interior Point Method

Solve:

$$\min\langle c, x\rangle \quad \text{s.t. } x \in \overline{dom f}.$$

Define:

$$f_\eta(x) = \eta\langle c, x\rangle + f(x).$$

Newton step:

$$n_\eta(x) = -H(x)^{-1}(\eta c + g(x)) = n(x) - \eta c.$$

Short-Step Method

Initialize:

$$\|n_{\eta_0}(x_0)\|_{x_0} \leq \frac{1}{9}.$$

Update:

$$\eta_{k+1} = \left(1 + \frac{1}{8\sqrt{\nu_f}}\right)\eta_k, \quad x_{k+1} = x_k + n_{\eta_{k+1}}(x_k).$$

Theorem 4.4.12. *If*

$$\|n_{\eta_k}(x_k)\|_{x_k} \leq \frac{1}{9},$$

then:

$$\|n_{\eta_{k+1}}(x_{k+1})\|_{x_{k+1}} \leq \frac{1}{9}.$$

Proof. Decompose:

$$n_{\eta_{k+1}}(x_k) = \frac{\eta_{k+1}}{\eta_k} n_{\eta_k}(x_k) + \left(1 - \frac{\eta_{k+1}}{\eta_k}\right) n(x_k),$$

and use SC bounds. □

Complexity

To reach $\eta_k \sim 1/\varepsilon$:

$$k = O(\sqrt{\nu_f}\log(1/\varepsilon)).$$

Alternative (Backward View)

Let $\tilde{c} = g(x)$.

Then x minimizes:

$$\min\langle \tilde{c}, x\rangle + f(x),$$

i.e., central path reversal.

Long-Step / Predictor-Corrector

Practical improvements:

- Long-step: increase η aggressively
- Multiple Newton corrections
- Predictor-corrector: jump + recenter

4.4.7 Primal-Dual View

Barrier problem:

$$\min f(x) + \langle c, x \rangle + B(x),$$

with:

$$\nabla B(x) = \sum_i \nabla g_i(x) \cdot \frac{1}{-g_i(x)}.$$

KKT-like system:

$$\begin{cases} \nabla f(x) + \nabla g(x)^T u + A^T y = 0 \\ u_i g_i(x) = -\frac{1}{\eta} \\ Ax = b \\ u_i > 0,\ g_i(x) < 0 \end{cases}$$

Thus interior-point methods can be viewed as:

Newton applied to perturbed KKT conditions.